Class #16 - LU Factorization with permutation

1. Example: Find an LU factorization of $A = \begin{bmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{bmatrix}$

2. Example: Find an LU factorization of $A = \begin{bmatrix} 0.0001 & 1 \\ 1 & 1 \end{bmatrix}$.

   (a) Is $LU = A$?

   (b) How to avoid such problem?

3. Def: Exchanging rows to obtain the largest possible pivot (in its absolute value) in that column is called *partial pivoting*.

   Example: The case #2(b) can be written as $PA = LU$ where $P =$ __________. Such factorization is called *permuted LU factorization*.

4. Def: A nonzero square matrix P is called a *permutation matrix* if there is exactly one nonzero entry in each row and column that is 1 and if the rest are all zero. This means, if $(\alpha_1, \alpha_2, \cdots, \alpha_n)$ is a permuation of $(1, 2, \cdots, n)$, then

$$P = \begin{bmatrix} {e_{\alpha_1}}^T \\ {e_{\alpha_2}}^T \\ \vdots \\ {e_{\alpha_n}}^T \end{bmatrix},$$

where e_i is the ith column of the $n \times n$ identity matrix.

5. Example: Let $(2, 3, 1)$ be a permution of $(1, 2, 3)$. Then $P =$ __________.

Let A be an arbitrary 3×3 matrix. What's PA?

How about AP?

6. Properties of Permutation matrices:

(a) The inverse of P is __________, which is also a __________matrix.

(b) The product of two permutation matrices is a __________matrix.